

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO5-ClassActivity

Programming in Java - File I/O Operations

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Course Code	CSA0904
Course Name	Programming in Java
Assignment Topic	CO5-ClassActivity: Copying Content Between Files
Question No.	Question 3

QUESTION 3: COPY CONTENT FROM SOURCE FILE TO DESTINATION FILE

Problem Statement:

Create a Java program to copy the entire contents of a source file (source.txt) to a destination file (destination.txt) using byte streams (FileInputStream, FileOutputStream, and buffer array) with proper stream management.

Source Code (CopyFileContent.java):

```
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class CopyFileContent {
    public static void main(String[] args) {
        String sourcePath = "source.txt";
        String destinationPath = "destination.txt";

        File sourceFile = new File(sourcePath);
        File destinationFile = new File(destinationPath);

        // Verify source file exists
        if (!sourceFile.exists()) {
            System.out.println("Error: Source file '" + sourcePath + "' does not exist.");
            return;
        }

        // Byte-stream copy (handles text, images, and binary files)
        try (FileInputStream inputStream = new FileInputStream(sourceFile);
            FileOutputStream outputStream = new FileOutputStream(destinationFile)) {

            byte[] buffer = new byte[1024];
            int bytesRead;
            long totalBytesCopied = 0;

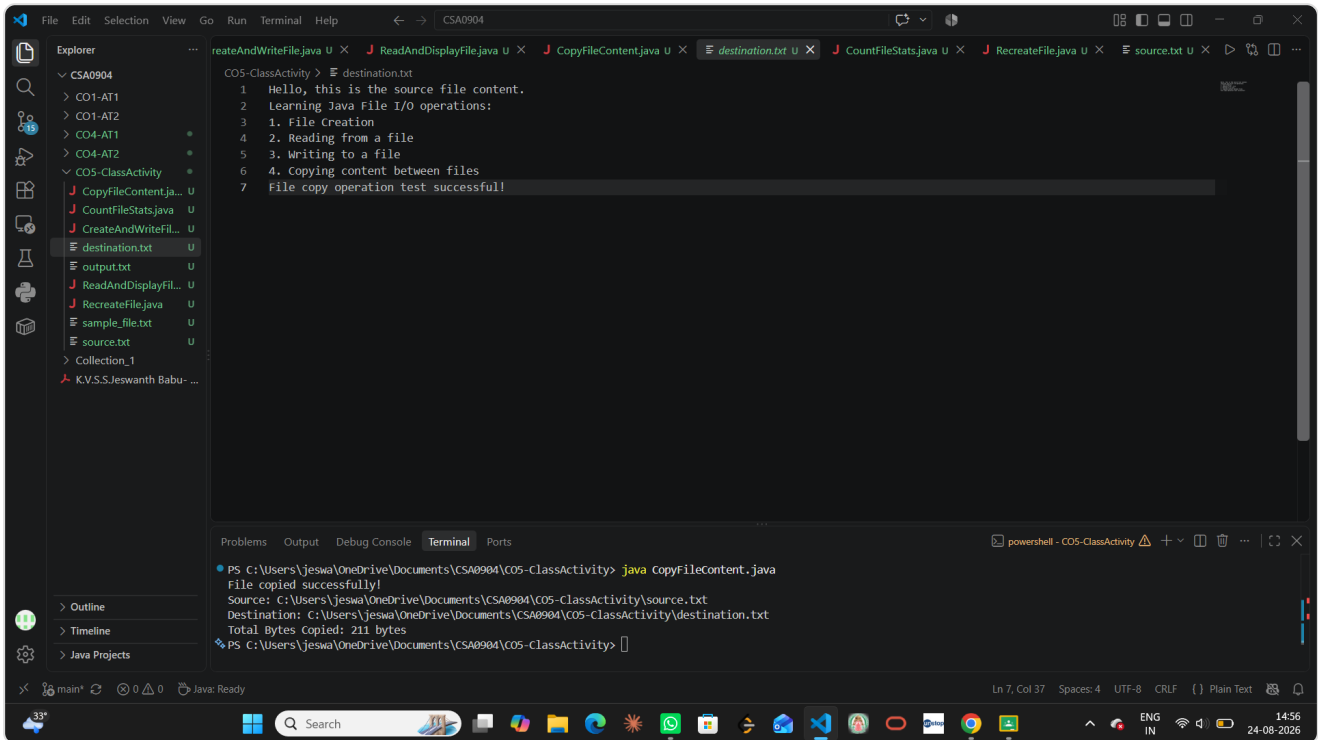
            while ((bytesRead = inputStream.read(buffer)) != -1) {
                outputStream.write(buffer, 0, bytesRead);
                totalBytesCopied += bytesRead;
            }

            System.out.println("File copied successfully!");
            System.out.println("Source: " + sourceFile.getAbsolutePath());
            System.out.println("Destination: " + destinationFile.getAbsolutePath());
            System.out.println("Total Bytes Copied: " + totalBytesCopied + " bytes");

        } catch (IOException e) {
            System.err.println("An error occurred during file copying: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

EXECUTION & VERIFICATION OUTPUT

1. Program Execution & Terminal Output:



The screenshot displays an IDE window with the following components:

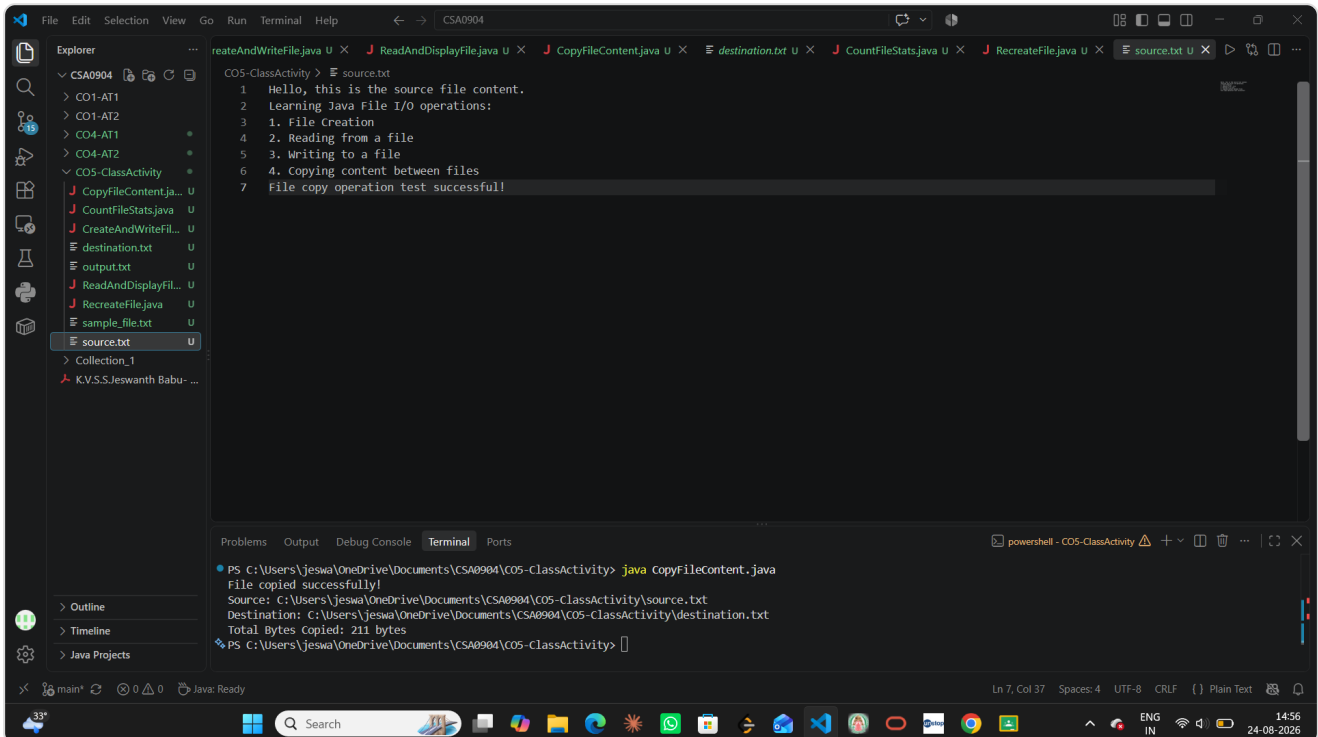
- Explorer:** Shows a project structure for 'CSA0904' with folders 'CO1-AT1', 'CO1-AT2', 'CO4-AT1', 'CO4-AT2', and 'CO5-ClassActivity'. The 'CO5-ClassActivity' folder contains files: 'CopyFileContent.java', 'CountFileStats.java', 'CreateAndWriteFil...', 'destination.txt', 'output.txt', 'ReadAndDisplayFil...', 'RecreateFile.java', 'sample_file.txt', 'source.txt', and 'Collection_1'.
- Editor:** Displays the content of 'destination.txt' with the following text:

```
CO5-ClassActivity > destination.txt
1 Hello, this is the source file content.
2 Learning Java File I/O operations:
3 1. File Creation
4 2. Reading from a file
5 3. Writing to a file
6 4. Copying content between files
7 File copy operation test successful!
```
- Terminal:** Shows the execution of the 'java copyFileContent.java' command. The output is:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java copyFileContent.java
File copied successfully!
Source: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\source.txt
Destination: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\destination.txt
Total Bytes Copied: 211 bytes
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity>
```

FILE VERIFICATION (SOURCE VS DESTINATION)

2. Original Source Content (source.txt):



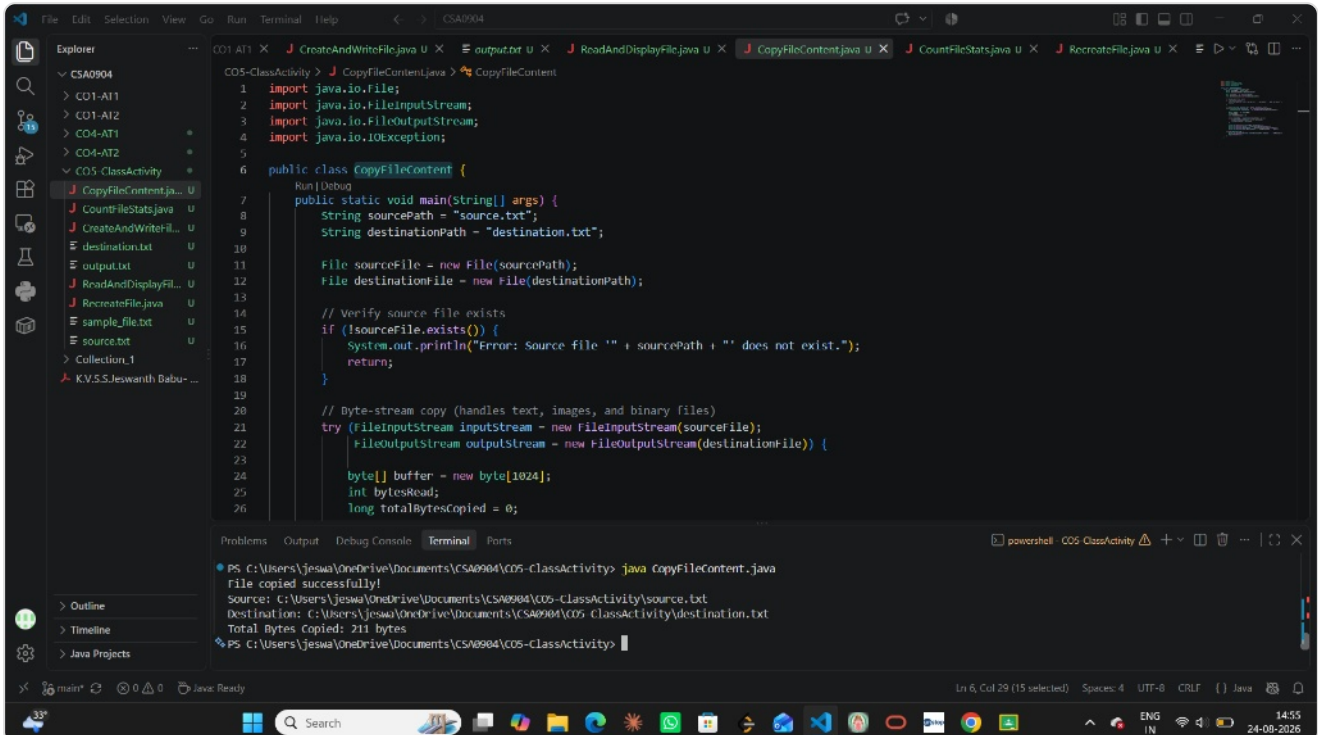
The screenshot shows an IDE with the Explorer view on the left displaying the project structure. The main editor shows the content of `source.txt`:

```
1 Hello, this is the source file content.
2 Learning Java File I/O operations:
3 1. File Creation
4 2. Reading from a file
5 3. Writing to a file
6 4. Copying content between files
7 File copy operation test successful!
```

The Terminal window at the bottom shows the command `java copyFileContent.java` being executed, with the following output:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java copyFileContent.java
File copied successfully!
Source: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\source.txt
Destination: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\destination.txt
Total Bytes Copied: 211 bytes
```

3. Successfully Copied Content (destination.txt):



The screenshot shows the same IDE with the `CopyFileContent.java` file open in the editor. The code is as follows:

```
1 import java.io.File;
2 import java.io.FileInputStream;
3 import java.io.FileOutputStream;
4 import java.io.IOException;
5
6 public class CopyFileContent {
7     public static void main(String[] args) {
8         String sourcePath = "source.txt";
9         String destinationPath = "destination.txt";
10
11         File sourceFile = new File(sourcePath);
12         File destinationFile = new File(destinationPath);
13
14         // Verify source file exists
15         if (!sourceFile.exists()) {
16             System.out.println("Error: Source file '" + sourcePath + "' does not exist.");
17             return;
18         }
19
20         // Byte-stream copy (handles text, images, and binary files)
21         try (FileInputStream inputStream = new FileInputStream(sourceFile);
22             FileOutputStream outputStream = new FileOutputStream(destinationFile)) {
23
24             byte[] buffer = new byte[1024];
25             int bytesRead;
26             long totalBytesCopied = 0;
```

The Terminal window shows the command `java CopyFileContent.java` being executed, with the following output:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java CopyFileContent.java
File copied successfully!
Source: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\source.txt
Destination: C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity\destination.txt
Total Bytes Copied: 211 bytes
```